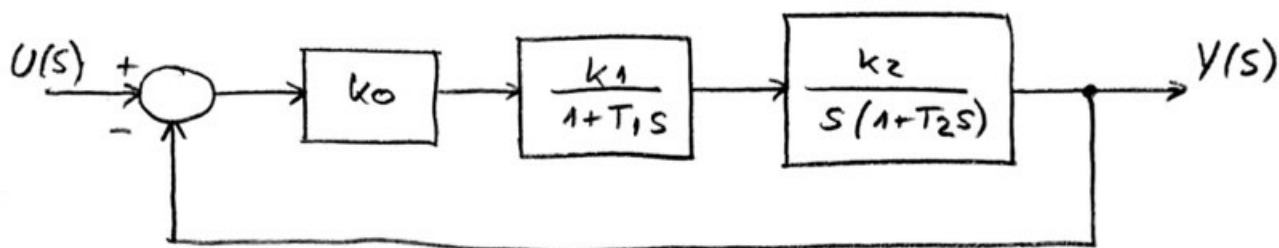


ZADATAK 1.

Odrediti koeficijent pojačanja otvorenog sistema ko prikazanog na slici tako da zadovolji Hurwitzow kriterij stabilnosti.

Zadano je: $T_1 = 0.025$, $T_2 = 0.16$ s, $k_1 = 6$ dB, $k_2 = 10$ dB



RJEŠENJE

$$G_0(s) = \frac{k_0 k_1 k_2}{s(1+T_1s)(1+T_2s)}$$

$$W(s) = \frac{G_0(s)}{1 + G_0(s)}$$

$$W(s) = \frac{\frac{k_0 k_1 k_2}{s(1+T_1s)(1+T_2s)}}{\frac{s(1+T_1s)(1+T_2s) + k_0 k_1 k_2}{s(1+T_1s)(1+T_2s)}}$$

$$W(s) = \frac{k_0 k_1 k_2}{\underbrace{s(1+T_1s)(1+T_2s) + k_0 k_1 k_2}_{(*)}}$$

$$K = k_0 k_1 k_2$$

$$(*) \quad s(1+T_1s)(1+T_2s) + K = 0$$

$$s(1+T_1s+T_2s+T_1T_2s^2) + K = 0$$

$$T_1T_2s^3 + (T_1+T_2)s^2 + s + K = 0$$

$$a_3 = T_1T_2$$

$$a_2 = T_1 + T_2$$

$$a_1 = 1$$

$$a_0 = K$$

$$n = 3$$

$$D_n = a_n D_{n-1} > 0$$

$$D_3 = a_3 D_2 > 0$$

$$D_2 = \begin{vmatrix} a_1 & a_0 \\ a_3 & a_2 \end{vmatrix} = \begin{vmatrix} 1 & K \\ T_1T_2 & T_1+T_2 \end{vmatrix} = T_1+T_2 - T_1T_2K > 0$$

$$-T_1T_2K > -(T_1+T_2) \quad /(-1)$$

$$K = k_0 k_1 k_2$$

$$K < \frac{T_1+T_2}{T_1T_2}$$

$$K < \underline{\underline{56.25}}$$

$$20 \log K = 20 \log k_0 + 20 \log k_1 + 20 \log k_2$$

$$20 \log k_0 = 20 \log K - 20 \log k_1 - 20 \log k_2$$

$$20 \log k_0 = 35 - 16 - 20$$

$$20 \log k_0 = -1$$

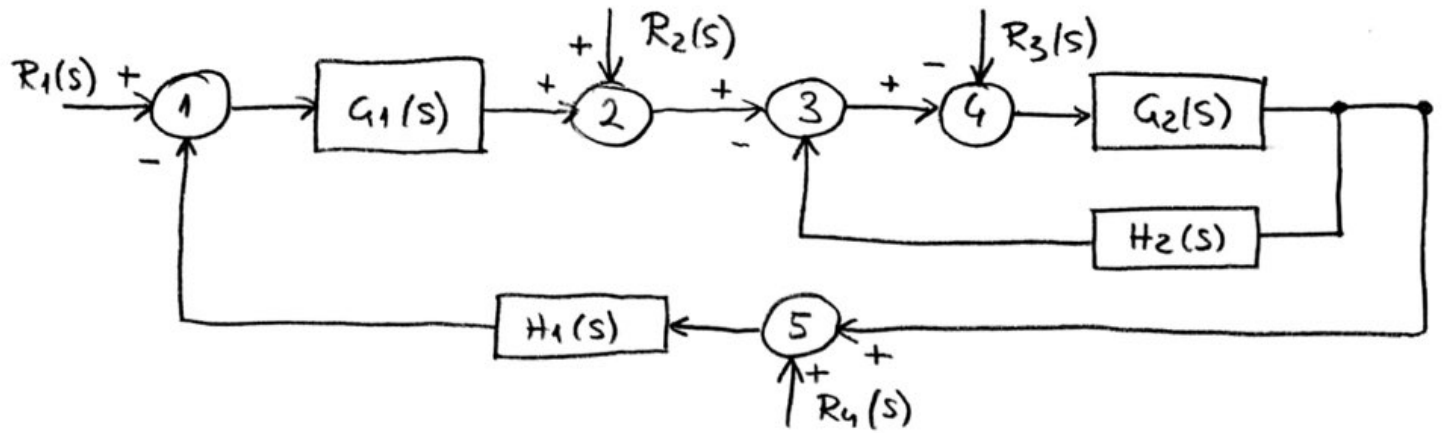
$$k_0 = 10^{-\frac{1}{20}}$$

$$\underline{\underline{k_0 = 0.8913}}$$

$$\underline{\underline{0 < K < 56.25}}$$

ZADATAK 1.

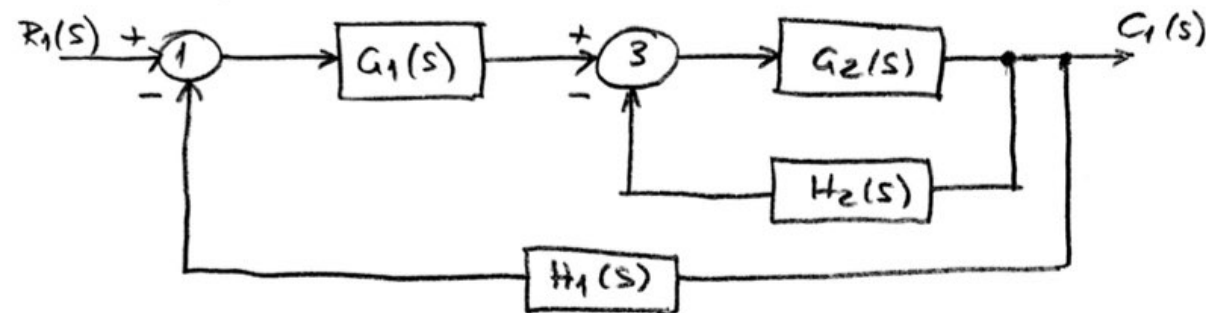
Primjenom pravila algebre blokova uprostiti dijagram sa slike:



RJEŠENJE

$$C(s) = C_1(s) + C_2(s) + C_3(s) + C_4(s)$$

$$C_1(s) = ?$$



$$G_2'(s) = \frac{G_2(s)}{1 + G_2(s)H_2(s)}$$

$$W_1(s) = ?$$

$$G_1'(s) = G_1(s) \cdot G_2'(s)$$

$$W_1(s) = \frac{G_1'(s)}{1 + G_1'(s)H_1(s)}$$

$$W_1(s) = \frac{C_1(s)}{R_1(s)}$$

$$W_1(s) = \frac{G_1(s) \cdot G_2'(s)}{1 + G_1(s) \cdot G_2'(s)H_1(s)}$$

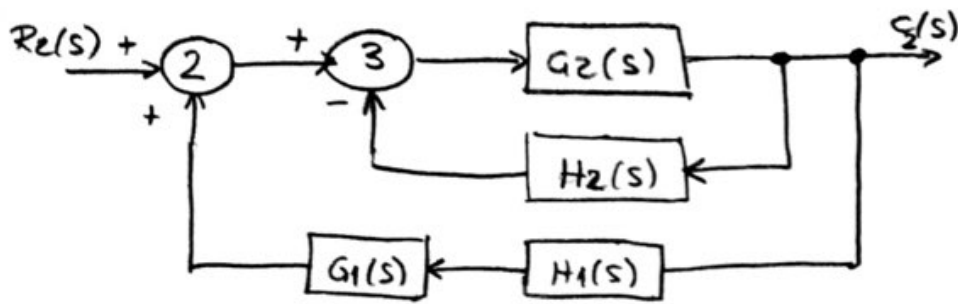
$$C_1(s) = W_1(s) \cdot R_1(s)$$

$$W_1(s) = \frac{\frac{G_1(s)G_2(s)}{1 + G_2(s)H_2(s)}}{1 + \frac{G_1(s)G_2(s)H_1(s)}{1 + G_2(s)H_2(s)}}$$

$$W_1(s) = \frac{G_1(s)G_2(s)}{1 + G_2(s)H_2(s) + G_1(s)G_2(s)H_1(s)}$$

$$C_1(s) = \frac{G_1(s)G_2(s)}{1 + G_2(s)H_2(s) + G_1(s)G_2(s)H_1(s)} \cdot R_1(s)$$

$$C_2(s) = ?$$



$$G_2'(s) = \frac{G_2(s)}{1 + G_2(s)H_2(s)}$$

$$W_2(s) = \frac{C_2(s)}{R_2(s)}$$

$$C_2(s) = W_2(s)R_2(s)$$

$$W_2(s) = ?$$

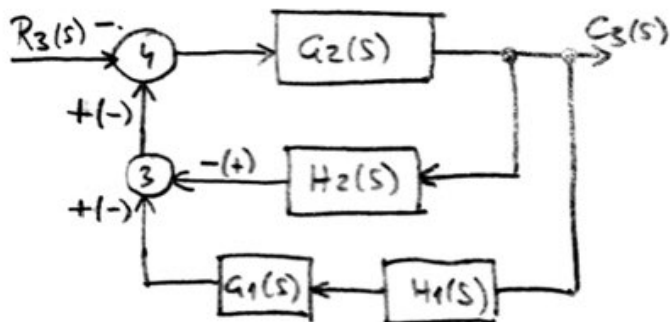
$$W_2(s) = \frac{G_2'(s)}{1 + G_2'(s)G_1(s)H_1(s)}$$

$$W_2(s) = \frac{\frac{G_2(s)}{1 + G_2(s)H_2(s)}}{1 + \frac{G_1(s)G_2(s)H_1(s)}{1 + G_2(s)H_2(s)}}$$

$$W_2(s) = \frac{G_2(s)}{1 + G_2(s)H_2(s) + G_1(s)G_2(s)H_1(s)}$$

$$C_2(s) = \frac{G_2(s)}{1 + G_2(s)H_2(s) + G_1(s)G_2(s)H_1(s)} \cdot R_2(s)$$

$$C_3(s) = ?$$



$$H'(s) = H_2(s) + G_1(s)H_1(s)$$

$$W_3 = \frac{C_3(s)}{-R_3(s)}$$

$$C_3(s) = -W_3(s)R_3(s)$$

$$W_3(s) = ?$$

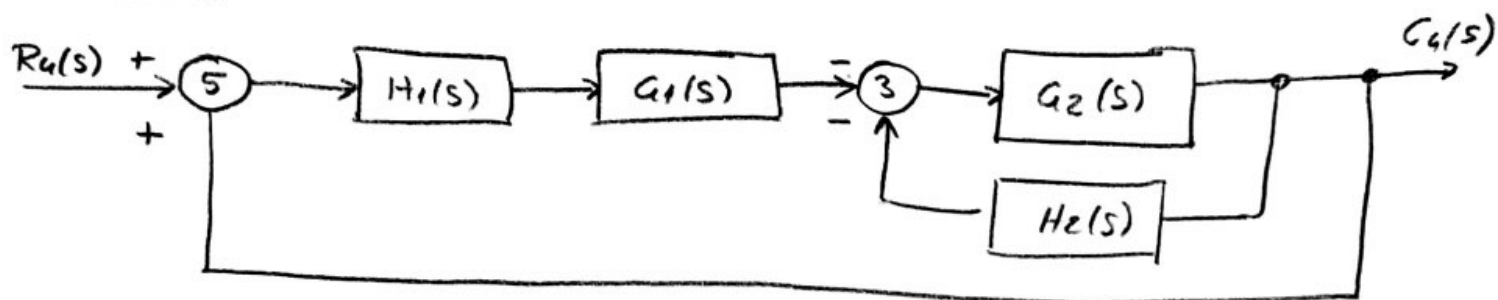
$$W_3(s) = \frac{G_2(s)}{1 + H'(s)G_2(s)}$$

$$W_3(s) = \frac{G_2(s)}{1 + G_2(s)(H_2(s) + G_1(s)H_1(s))}$$

$$W_3(s) = \frac{G_2(s)}{1 + G_2(s)H_2(s) + G_1(s)G_2(s)H_1(s)}$$

$$C_3(s) = - \frac{G_2(s)}{1 + G_2(s)H_2(s) + G_1(s)G_2(s)H_1(s)} \cdot R_3(s)$$

$$-G_2'(s) = ?$$



$$G_2'(s) = \frac{G_2(s)}{1 + G_2(s)H_2(s)}$$

$$W_4(s) = ?$$

$$W_4(s) = \frac{H_1(s)G_1(s)(-G_2'(s))}{1 - H_1(s)G_1(s)(-G_2'(s))}$$

$$W_4(s) = \frac{C_4(s)}{R_4(s)}$$

$$C_4(s) = W_4(s)R_4(s)$$

$$W_4(s) = - \frac{H_1(s)G_1(s)G_2'(s)}{1 + H_1(s)G_1(s)G_2'(s)}$$

$$W_4(s) = - \frac{H_1(s)G_1(s) \frac{G_2(s)}{1 + G_2(s)H_2(s)}}{1 + H_1(s)G_1(s) \frac{G_2(s)}{1 + G_2(s)H_2(s)}}$$

$$W_4(s) = - \frac{H_1(s)G_1(s)G_2(s)}{1 + G_2(s)H_2(s) + G_1(s)G_2(s)H_1(s)}$$

$$C_4(s) = - \frac{G_1(s)G_2(s)H_1(s)}{1 + G_2(s)H_2(s) + G_1(s)G_2(s)H_1(s)} \cdot R_4(s)$$

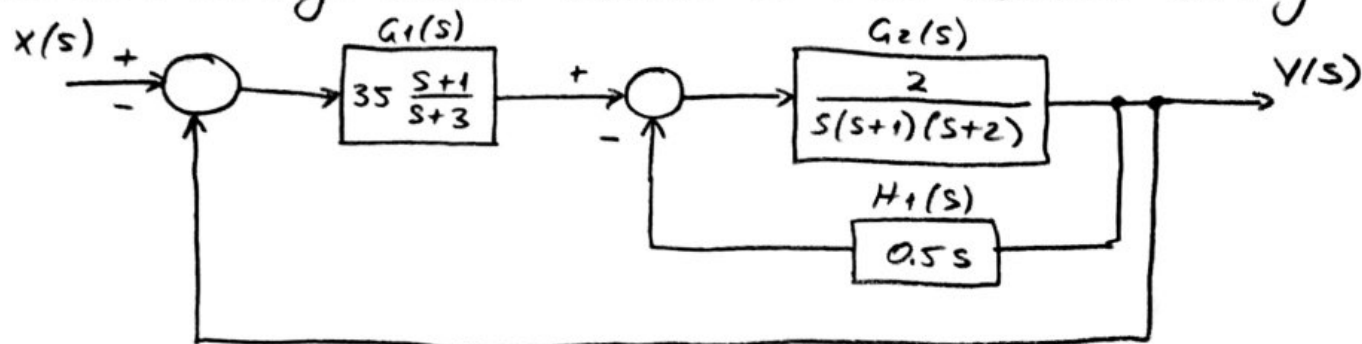
$$C(s) = C_1(s) + C_2(s) + C_3(s) + C_4(s)$$

$$C(s) = \frac{G_1(s)G_2(s)R_1(s) + G_2(s)R_2(s) - G_2(s)R_3(s) - G_1(s)G_2(s)H_1(s)R_4(s)}{1 + G_2(s)H_2(s) + G_1(s)G_2(s)H_1(s)}$$

$$C(s) = \frac{G_1(s)G_2(s)R_1(s) + G_2(s)[R_2(s) - R_3(s)] - G_1(s)G_2(s)H_1(s)R_4(s)}{1 + G_2(s)H_2(s) + G_1(s)G_2(s)H_1(s)}$$

INER 2.

Odrediti da li je sistem stabilan koristeći Routhov kriterij.



RJEŠENJE

$$G_2'(s) = \frac{G_2(s)}{1 + G_2(s)H_1(s)}$$

$$W(s) = \frac{Y(s)}{X(s)} = \frac{G_1(s)G_2'(s)}{1 + G_1(s)G_2'(s)}$$

$$W(s) = \frac{\frac{G_1(s)G_2(s)}{1 + G_2(s)H_1(s)}}{1 + \frac{G_1(s)G_2(s)}{1 + G_2(s)H_1(s)}}$$

$$W(s) = \frac{G_1(s)G_2(s)}{1 + G_1(s)G_2(s) + G_2(s)H_1(s)}$$

s^4	1	12	70
s^3	6	79	0
s^2	b_1	b_2	b_3
s^1	c_1	c_2	c_3
s^0	70		

$$b_1 = \frac{6 \cdot 12 - 1 \cdot 79}{6} = \frac{72 - 79}{6} = -\frac{7}{6}$$

$$b_2 = \frac{6 \cdot 70 - 1 \cdot 0}{6} = \frac{420}{6} = 70$$

$$c_1 = \frac{-\frac{7}{6} \cdot 79 - 70 \cdot 6}{-\frac{7}{6}} = \frac{-\frac{553}{6} - 420}{-\frac{7}{6}} = +439$$

s^4	1
s^3	6
s^2	$-\frac{7}{6}$
s^1	439
s^0	70

Sistem je nestabilan!

$$W(s) = \frac{\frac{70(s+1)}{s(s+1)(s+2)(s+3)}}{1 + \frac{70(s+1)}{s(s+1)(s+2)(s+3)} + \frac{s}{s(s+1)(s+2)}}$$

$$W(s) = \frac{70(s+1)}{s(s+1)(s+2)(s+3) + 70(s+1) + s(s+3)}$$

$$f(s) = s(s+1)(s+2)(s+3) + 70(s+1) + s(s+3)$$

$$f(s) = (s^2+s)(s^2+2s+3s+6) + 70s+70 + s^2+3s$$

$$f(s) = s^4 + s^3 + 2s^3 + 2s^2 + 3s^3 + 3s^2 + 6s^2 + 6s + 70s + 70 + s^2 + 3s$$

$$f(s) = s^4 + 6s^3 + 12s^2 + 43s + 70$$

$$a_4 = 1 \quad a_3 = 6 \quad a_2 = 12 \quad a_1 = 43 \quad a_0 = 70$$

DATAK 3.

zadana je prijenosna funkcija duoreenog sistema. Potrebno je:

- Hurwitzovom metodom odrediti da li je sistem stabilan
- nacrtati Bodeove dijagrame i zaključiti o stabilnosti sistema

$$W_0(s) = \frac{20}{s(s+2)(s+5)}$$

a) Hurwitzov kriterij

$$W(s) = \frac{W_0(s)}{1+W_0(s)} \Rightarrow W(s) = \frac{20}{s(s+2)(s+5)+20}$$

$$f(s) = (s^2+2s)(s+5)+20$$

$$f(s) = s^3 + 2s^2 + 5s^2 + 10s + 20$$

$$f(s) = s^3 + 7s^2 + 10s + 20$$

$$\underline{a_3 = 1 \quad a_2 = 7 \quad a_1 = 10 \quad a_0 = 20}$$

$$\Delta_1 = a_{n-1} \Rightarrow \Delta_1 = 7 > 0$$

$$\Delta_2 = \begin{vmatrix} a_{n-1} & a_{n-3} \\ a_n & a_{n-2} \end{vmatrix} \Rightarrow \Delta_2 = \begin{vmatrix} 7 & 20 \\ 1 & 10 \end{vmatrix} \Rightarrow \Delta_2 = 50 > 0$$

$$\Delta_3 = \begin{vmatrix} a_{n-1} & a_{n-3} & a_{n-5} \\ a_n & a_{n-2} & a_{n-4} \\ 0 & a_{n-1} & a_{n-3} \end{vmatrix} \Rightarrow \Delta_3 = \begin{vmatrix} 7 & 20 & 0 \\ 1 & 10 & 0 \\ 0 & 7 & 20 \end{vmatrix} \Rightarrow \Delta_3 = 7 \begin{vmatrix} 10 & 0 \\ 7 & 20 \end{vmatrix} - 20 \begin{vmatrix} 1 & 0 \\ 0 & 20 \end{vmatrix}$$

$$\Delta_3 = 7 \cdot 200 - 20 \cdot 20 \Rightarrow \Delta_3 = 1000 > 0$$

$$\Delta_1 > 0 \quad \Delta_2 > 0 \quad \Delta_3 > 0$$

Sistem je stabilan!

b) Bodeov kriterij stabilnosti

$$W_0(s) = \frac{20}{s(s+2)(s+5)}$$

$$W_0(s) = \frac{20^2}{s \cdot 2(1+\frac{s}{2}) \cdot 5(1+\frac{s}{5})}$$

$$W_0(s) = \frac{2}{s(1+\frac{s}{2})(1+\frac{s}{5})}$$

$$W_0(j\omega) = \frac{2}{j\omega(1+j\frac{\omega}{2})(1+j\frac{\omega}{5})}$$

$$A = 2$$

$$|A| = 20 \log |2| \text{ dB} = 6 \text{ dB}$$

$$\varphi_A = \arctg \frac{0}{10} = 0^\circ$$

$$B = \frac{1}{j\omega}$$

$$|B| = -20 \log |\omega| \text{ dB}$$

$$\varphi_B = -\arctg \frac{\omega}{0} = -90^\circ$$

$$C = \frac{1}{1+j\frac{\omega}{2}}$$

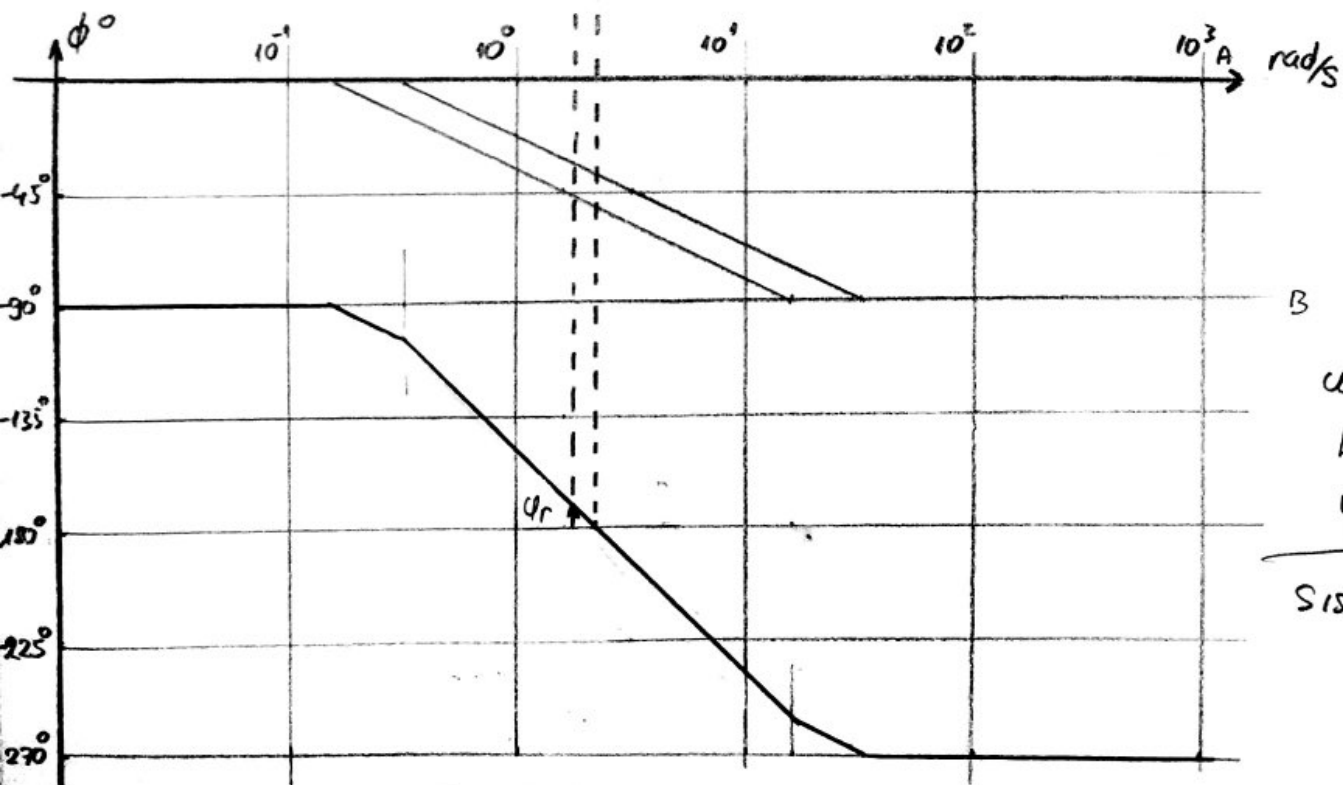
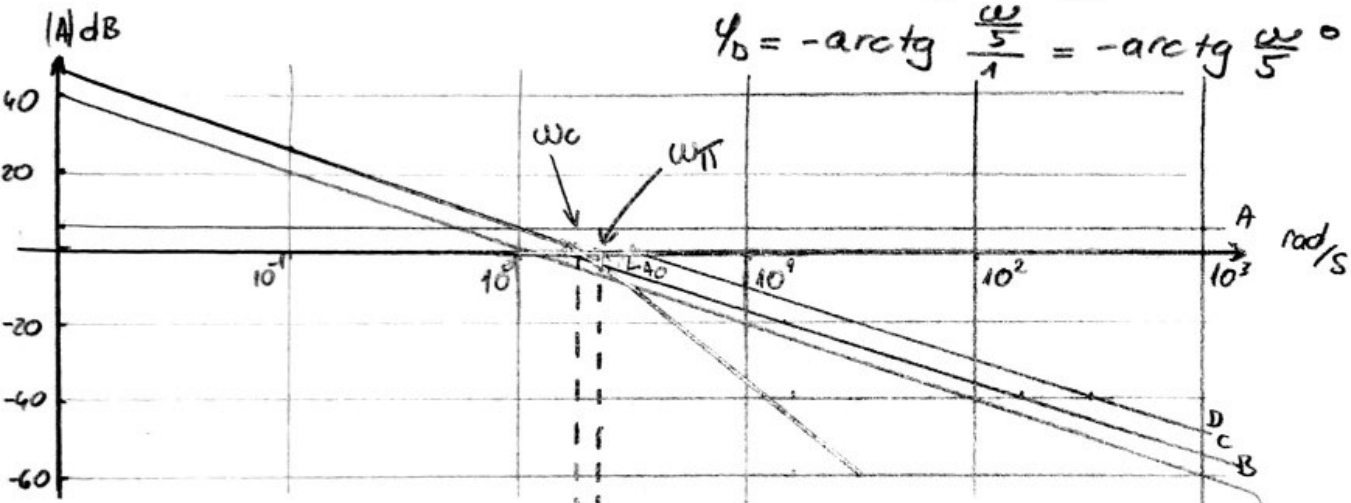
$$|C| = -20 \log \sqrt{1+(\frac{\omega}{2})^2} \text{ dB}$$

$$\varphi_C = -\arctg \frac{\frac{\omega}{2}}{1} = -\arctg \frac{\omega}{2}^\circ$$

$$D = \frac{1}{1+j\frac{\omega}{5}}$$

$$|D| = -20 \log \sqrt{1+(\frac{\omega}{5})^2}$$

$$\varphi_D = -\arctg \frac{\frac{\omega}{5}}{1} = -\arctg \frac{\omega}{5}^\circ$$



$$\omega_c < \omega_{\pi} \checkmark$$

$$L_{AO} > 0$$

$$\varphi_r > 0$$

Sistem stabilan

ZADATAK 1.

Na sistem zadane prijenosnom funkcijom:

$$G(s) = \frac{Y(s)}{X(s)} = \frac{2-s}{(s+1)(s+4)}$$

djeluje pobuda:

$$u(t) = 10(e^{-t} - e^{-5t}) \cdot s(t)$$

Potrebno je:

a) Odrediti diferencijalnu jednačinu koja opisuje ponašanje sistema

b) Odrediti odziv $y(t)$ za $t > 0$ ako je poznato $y(0) = 1$ i $y'(0) = -1$

RJEŠENJE

$$a) \quad \frac{Y(s)}{X(s)} = \frac{2-s}{(s+1)(s+4)} \quad | \cdot X(s)(s+1)(s+4)$$

$$Y(s) [(s+1)(s+4)] = (2-s) X(s)$$

$$Y(s) [s^2 + 5s + 4] = 2X(s) - sX(s)$$

$$s^2 Y(s) + 5s Y(s) + 4Y(s) = -sX(s) + 2X(s) \quad | \cdot \mathcal{L}^{-1}$$

$$y'' + 5y' + 4y = -x' + 2x$$

$$y''(t) + 5y'(t) + 4y(t) = 2x(t) - x'(t)$$

$$b) \quad s^2 Y(s) - s + 1 + 5s Y(s) - 5 + 4Y(s) = 2X(s) - sX(s)$$

$$Y(s) [s^2 + 5s + 4] - s - 4 = X(s) (2-s)$$

$$s^2 + 4s + 4s + 4 = s(s+1) + 4(s+1) = (s+1)(s+4)$$

$$Y(s) = \frac{s+4}{(s+1)(s+4)} + \frac{2-s}{(s+1)(s+4)} \cdot X(s)$$

$$u(s) = X(s) = \frac{10}{s+1} - \frac{10}{s+5} = \frac{10(s+5) - 10(s+1)}{(s+1)(s+5)}$$

$$u(s) = \frac{10s + 50 - 10s - 10}{(s+1)(s+5)}$$

$$u(s) = \frac{40}{(s+1)(s+5)}$$

$$y(s) = \frac{s+4}{(s+1)(s+4)} + \frac{2-s}{(s+1)(s+4)} \cdot \frac{40}{(s+1)(s+5)}$$

$$= (s+4)/(s+1)(s+5)$$

$$y(s) = \frac{(s+4)(s+1)(s+5) + (2-s) \cdot 40}{(s+1)^2(s+4)(s+5)}$$

$$= (s^2 + 5s + 4)/(s+5)$$

$$= s^3 + 5s^2 + 4s + 5s^2 + 25s + 20$$

$$y(s) = \frac{s^3 + 10s^2 + 29s + 20 + 80 - 40s}{(s+1)^2(s+4)(s+5)}$$

$$= s^3 + 10s^2 + 29s + 20$$

$$y(s) = \frac{s^3 + 10s^2 - 11s + 100}{(s+1)^2(s+4)(s+5)}$$

$$\frac{s^3 + 10s^2 - 11s + 100}{(s+1)^2(s+4)(s+5)} = \frac{A}{(s+1)} + \frac{B}{(s+1)^2} + \frac{C}{(s+4)} + \frac{D}{(s+5)} \quad | \cdot (s+1)^2(s+4)(s+5)$$

$$s^3 + 10s^2 - 11s + 100 = A(s+1)(s+4)(s+5) + B(s+4)(s+5) + C(s+1)^2(s+5) + D(s+1)^2(s+4)$$

$$s^3 + 10s^2 - 11s + 100 = A(s^3 + 10s^2 + 29s + 20) + B(s^2 + 9s + 20)$$

$$+ C(s^3 + 7s^2 + 11s + 5) + D(s^3 + 6s^2 + 9s + 4)$$

$$A + C + D = 1 \quad A = 1 - C - D$$

$$10A + B + 7C + 6D = 10$$

$$29A + 9B + 11C + 9D = -11$$

$$20A + 20B + 5C + 4D = 100$$

$$10 - 10C - 10D + B + 7C + 6D = 10$$

$$29 - 29C - 29D + 9B + 11C + 9D = -11$$

$$20 - 20C - 20D + 20B + 5C + 4D = 100$$

$$B - 3C - 4D = 0$$

$$9B - 18C - 20D = -40$$

$$20B - 15C - 16D = 80$$

Determinanta

$$B = \frac{D_B}{D} = \frac{-1440}{-144} = 10$$

$$C = \frac{D_C}{D} = \frac{-3840}{-144} = \frac{80}{3}$$

$$D = \frac{D_D}{D} = \frac{2520}{-144} = -\frac{35}{2}$$

$$A = 1 - \frac{80}{3} + \frac{35}{2} = \frac{6 - 160 + 105}{6} = -\frac{49}{6}$$

$$y(s) = \frac{-\frac{49}{6}}{s+1} + \frac{10}{(s+1)^2} + \frac{\frac{80}{3}}{s+4} - \frac{\frac{35}{2}}{s+5}$$

$$y(t) = -\frac{49}{6} e^{-t} + 10te^{-t} + \frac{80}{3} e^{-4t} - \frac{35}{2} e^{-5t}$$

$$D = \begin{vmatrix} 1 & -3 & -4 \\ 9 & -18 & -20 \\ 20 & -15 & -16 \end{vmatrix} = -12 + 768 - 3000$$

$$D = -144$$

$$D_B = \begin{vmatrix} 0 & -3 & -4 \\ -40 & -18 & -20 \\ 80 & -15 & -16 \end{vmatrix} = 6720 - 8160$$

$$D_B = -1440$$

$$D_C = \begin{vmatrix} 1 & 0 & -4 \\ 9 & -40 & -20 \\ 20 & 80 & -16 \end{vmatrix} = 2240 - 6080$$

$$D_C = -3840$$

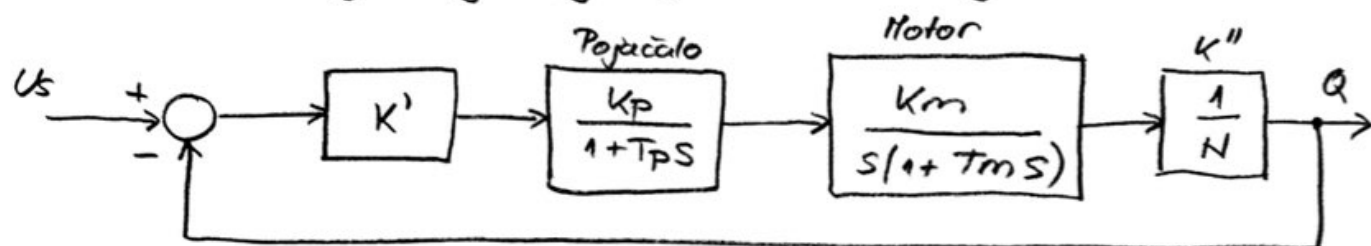
$$D_D = \begin{vmatrix} 1 & -3 & 0 \\ 9 & -18 & -40 \\ 20 & -15 & 80 \end{vmatrix} = -2040 + 4560$$

$$D_D = 2520$$

ATAK 2.

Za zadani sistem potrebno je:

- a) pomoću Hurwitzovog kriterija odrediti iznos pojačanja K za koji je sistem stabilan
- b) Za unijednosti pojačanja $K=80$ nacrtati Bodeove dijagrame i zaključiti o relativnoj stabilnosti
- c) Odrediti za koje pojačanje je sistem na granici stabilnosti



Poznato je: $T_p = 0.02 \text{ s}$, $T_m = 0.16 \text{ s}$

RIJEŠENJE

$$W(s) = \frac{Q(s)}{U_s(s)} = \frac{\frac{K}{s(1+T_p s)(1+T_m s)}}{1 + \frac{K}{s(1+T_p s)(1+T_m s)}} \Rightarrow W(s) = \frac{K}{s(1+T_p s)(1+T_m s) + K}$$

$$(*) F(s) = s(1+T_p s)(1+T_m s) + K$$

$$F(s) = s(1 + T_p s + T_m s + T_p T_m s^2) + K$$

$$F(s) = T_p T_m s^3 + (T_p + T_m)s^2 + s + K$$

$$F(s) = 0.0032 s^3 + 0.18 s^2 + s + K$$

$$a_3 = 0.0032 \quad a_2 = 0.18 \quad a_1 = 1 \quad a_0 = K$$

$$\Delta_1 = a_{n-1} \Rightarrow \Delta_1 = 0.18$$

$$\Delta_2 = \begin{vmatrix} a_{n-1} & a_{n-3} \\ a_n & a_{n-2} \end{vmatrix} \Rightarrow \Delta_2 = \begin{vmatrix} 0.18 & K \\ 0.0032 & 1 \end{vmatrix} \Rightarrow \Delta_2 = 0.18 - 0.0032 K > 0$$

$$K < 56.25$$

$$\Delta_3 = \begin{vmatrix} a_{n-1} & a_{n-3} & 0 \\ a_n & a_{n-2} & 0 \\ 0 & a_{n-1} & a_{n-3} \end{vmatrix} \Rightarrow \Delta_3 = \begin{vmatrix} 0.18 & K & 0 \\ 0.0032 & 1 & 0 \\ 0 & 0.18 & K \end{vmatrix} \Rightarrow \Delta_3 = 0.18 K - 0.0032 K^2 > 0$$

$$K(0.18 - 0.0032 K) > 0$$

$$\text{I } K > 0$$

$$\text{II } K < 56.25$$

Sistem je stabilan za:

$$0 < K < 56.25$$

$$W_0(s) = \frac{80}{s(1+0.02s)(1+0.16s)}$$

$$W_0(j\omega) = \frac{80}{j\omega(1+\frac{j\omega}{50})(1+\frac{4j\omega}{25})}$$

$$W_0(j\omega) = \frac{80}{j\omega(1+\frac{j\omega}{50})(1+\frac{j\omega}{6.25})}$$

$$A = 80$$

$$|A| = 20 \log |80| \text{ dB} = 38 \text{ dB}$$

$$\varphi = \arctg \frac{0}{80} = 0^\circ$$

$$B = \frac{1}{j\omega}$$

$$|A| = -20 \log(\omega) \text{ dB}$$

$$\varphi = -\arctg \frac{\omega}{0} = -90^\circ$$

$$C = \frac{1}{1+\frac{j\omega}{50}}$$

$$|A| = -20 \log \sqrt{1+(\frac{\omega}{50})^2} \text{ dB}$$

$$\varphi = -20 \arctg \frac{\omega}{50} = -\arctg \frac{\omega}{50}^\circ$$

$$D = \frac{1}{1+\frac{j\omega}{6.25}}$$

$$|A| = -20 \log \sqrt{1+(\frac{\omega}{6.25})^2} \text{ dB}$$

$$\varphi = -20 \arctg \frac{\omega}{6.25} = -\arctg \frac{\omega}{6.25}^\circ$$

Podaci po skripti

$$L_{A0} < 0$$

$$L_{A0} \approx -3 \text{ dB}$$

$$\varphi_r < 0$$

$$\varphi_r \approx -9^\circ$$

$$\omega_\pi < \omega_c$$

$$\omega_\pi = 18.5$$

$$\omega_c \approx 22$$

Sistem je relativno nestabilan.

$$c) L_{A0} = -3 \text{ dB}$$

$$20 \log(x) = 3 \text{ dB}$$

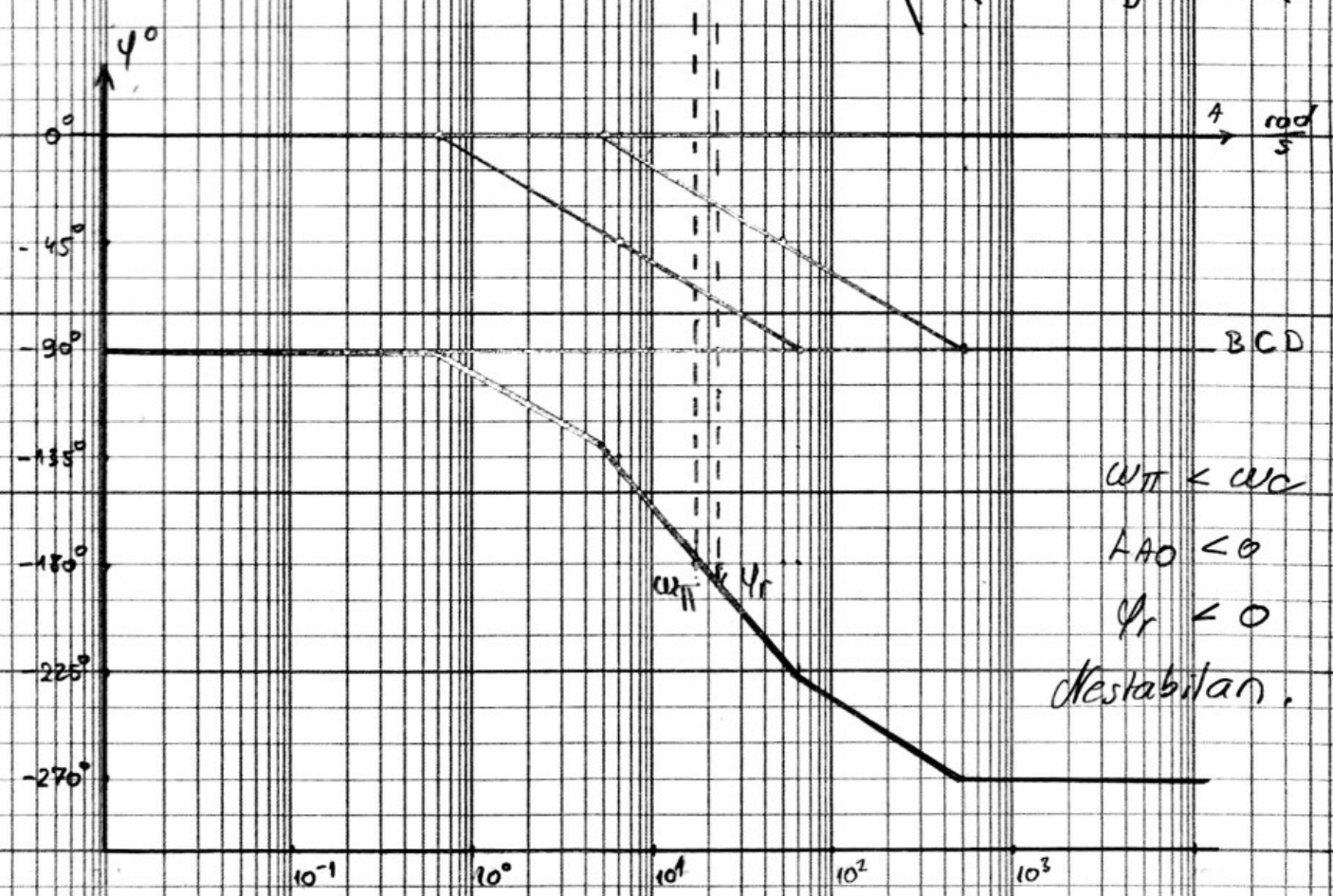
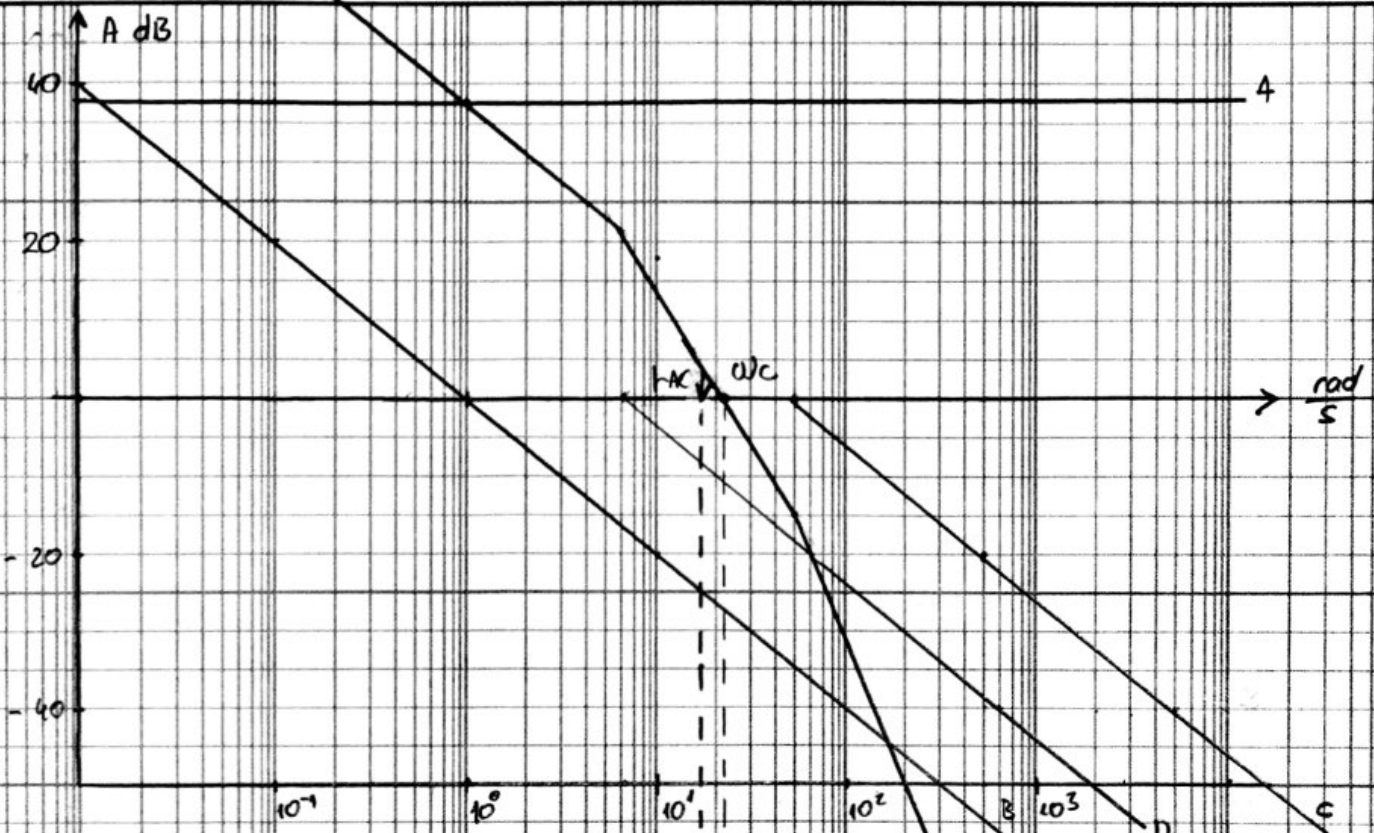
$$x = 10^{\frac{3}{20}}$$

$$x = 1.4125$$

Pojacanje 80 treba smanjiti 1.4125 puta da bi sistem postao granicno stabilan, odnosno

$$\frac{80}{1.4125} = 56.636$$

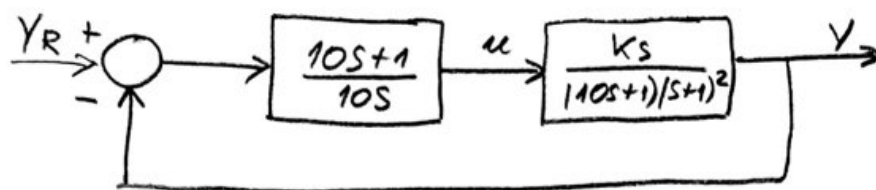
Pojacanje A treba biti 56.636.



$\omega_{\pi} < \omega_c$
 $A_G < 0$
 $\varphi_r < 0$
 nestabilan.

DATAK 3.

Za sistem koji je zadan na slici nacrtati GPK kada se k mijenja od nule do beskonačno.



RJEŠENJE

$$G_0(s) = K_S \frac{10s+1}{10s(10s+1)(s+1)^2} \Rightarrow G_0(s) = K_0 \frac{s + \frac{1}{10}}{s(s + \frac{1}{10})(s+1)^2}; \quad K_0 = \frac{K_S}{10} > 0$$

1° NULE: $m=1 \quad S_{N1} = -\frac{1}{10}$

2° POLOVI: $n=4 \quad SP_1=0 \quad SP_2=-\frac{1}{10} \quad SP_3=-1 \quad SP_4=-1$

3° $S = \sigma + j\omega \quad 4^\circ n-m=3$
 $\sigma \in (-\infty, 0]$

5° $\zeta_a = \frac{1}{n-m} \left\{ \sum_{v=1}^n \operatorname{Re}(SP_v) - \sum_{\mu=1}^m \operatorname{Re}(SN_\mu) \right\}$

$$\zeta_a = \frac{1}{3} \left\{ \left(0 - \frac{1}{10} - 1 - 1 \right) - \left(-\frac{1}{10} \right) \right\}$$

$$\zeta_a = \frac{1}{3} \left(-\frac{21}{10} + \frac{1}{10} \right)$$

$$\zeta_a = -\frac{2}{3}$$

6° $\alpha_k = \frac{180(2k-1)}{n-m} \quad k=1, 2, \dots, n-m$

$$\alpha_1 = \frac{180}{3} = 60^\circ$$

$$\alpha_2 = \frac{540}{3} = 180^\circ$$

$$\alpha_3 = \frac{900}{3} = 300^\circ = -60^\circ$$

7° $S = \zeta_v$

$$\sum_{v=1}^n \frac{1}{s - SP_v} = \sum_{\mu=1}^m \frac{1}{s - SN_\mu}$$

$$\frac{1}{s} + \frac{1}{s + \frac{1}{10}} + \frac{1}{s+1} + \frac{1}{s+1} = \frac{1}{s + \frac{1}{10}} \quad | \cdot s(s + \frac{1}{10})(s+1)$$

$$\vdots$$

$$s = -\frac{1}{3}$$

$$\zeta_v = -\frac{1}{3} \in \text{GPK}$$

